MSCR

Quiz: bit.ly/4eGheh6

1. What is the primary purpose of scientific research?

A) To prove personal opinions
B) To generate reliable, valid knowledge
C) To support existing beliefs
D) To entertain the public

2. Which is a characteristic of a good research problem?

A) Ambiguity
B) Simplicity
C) Specificity
D) Subjectivity

3. What is the first step in the scientific research process?

A) Conducting experiments
B) Reviewing the literature
C) Formulating a hypothesis
D) Analyzing data

4. Which best describes a hypothesis?

A) A proven fact
B) A testable statement
C) A random guess
D) An experimental result

5. What is the primary purpose of a literature review?

A) To gather raw data
B) To identify gaps in existing research
C) To formulate final hypotheses
D) To draw conclusions

6. Which method gathers observable, empirical, and measurable evidence systematically?

A) Qualitative research
B) Quantitative research
C) Theoretical research
D) Philosophical research

7. What is an independent variable?

A) A variable that is measured
B) A variable that is manipulated
C) A variable kept constant
D) A variable dependent on others

8. Which is NOT a characteristic of qualitative research?

A) Subjective analysis
B) Use of open-ended questions
C) Statistical analysis
D) Exploratory nature

9. Which sampling method divides the population into subgroups and samples each?

A) Simple random sampling
B) Systematic sampling
C) Stratified sampling
D) Convenience sampling

10. The consistency of a study or test is called:

A) Validity
B) Reliability
C) Accuracy
D) Precision

11. What is a control group?

A) The group that receives the treatment
B) The group used to measure the treatment effect
C) A group not in the study
D) The group that designs the experiment

12. Which is a qualitative data collection method?

A) Surveys
B) Experiments
C) Interviews
D) Quasi-experiments

13. In research, 'bias' refers to:

A) The precision of measurements
B) A systematic error that skews results
C) A random error
D) The validity of an experiment

14. What is the purpose of a double-blind study?

A) To increase sample size
B) To prevent bias
C) To decrease cost
D) To simplify procedures

15. What is 'operationalization'?

A) Defining variables in practical, measurable terms
B) Conducting the research
C) Analyzing data
D) Reviewing literature

16. 'Statistical significance' implies that:

A) Results are due to chance
B) Results are important
C) Results have practical significance
D) Results are unlikely due to chance

17. Which design best determines causality?

A) Descriptive
B) Correlational
C) Experimental
D) Exploratory

18. NOT a quantitative data collection method:

A) Questionnaires
B) Structured interviews
C) Focus groups
D) Surveys

19. Main goal of a pilot study:

A) Complete the full study
B) Test feasibility of design
C) Analyze data
D) Review literature

20. Research that seeks to understand phenomena in natural settings

A) Quantitative
B) Qualitative
C) Experimental
D) Theoretical

21. A case study is:

A) A study with a large sample
B) An in-depth analysis of a single/few cases
C) A study using only statistics
D) Purely theoretical

22. The extent to which results generalize to other people/settings:

A) Internal validity
B) External validity
C) Construct validity
D) Content validity

23. Exploratory research mainly:

A) Tests specific hypotheses
B) Establishes causality
C) Explores under-researched areas
D) Uses large samples

24. In experiments, which variable is influenced by the independent variable?

A) Control variable
B) Dependent variable
C) Extraneous variable
D) Confounding variable

25. Triangulation in qualitative research is:

A) Using multiple data sources/methods to verify findings
B) Using statistics to validate results
C) Using control groups
D) Testing hypotheses experimentally

26. A common qualitative data analysis method:

A) Regression analysis
B) Content analysis
C) ANOVA
D) Chi-square test

27. Intensive study of an individual/group/event:

A) Survey research
B) Experimental research
C) Case study research
D) Cross-sectional research

28. NOT an ethical consideration:

A) Informed consent
B) Confidentiality
C) Subjectivity
D) Avoiding harm

29. Main advantage of longitudinal studies:

A) Less expensive
B) Smaller samples
C) Study changes over time
D) Quick results

30. 'Peer review' means:

A) Review by friends/colleagues
B) Evaluation by field experts
C) Review by participants
D) Evaluation by funders

31. Research that manipulates variables to establish cause–effect:

A) Descriptive
B) Experimental
C) Correlational
D) Qualitative

32. Primary purpose of the null hypothesis:

A) Predict a specific outcome
B) Be proven true
C) Be tested and possibly rejected
D) Support the research hypothesis

33. Mixed-methods research:

A) Only quantitative
B) Only qualitative
C) Combines qualitative and quantitative
D) Uses only secondary data

34. Key characteristic of action research:

A) Controlled experiment
B) Solves an immediate, practical problem
C) Focus on theory only
D) Large samples

35. Ability of a test to measure what it should:

A) Reliability
B) Validity
C) Objectivity
D) Subjectivity

36. NOT a type of observational research:

A) Naturalistic observation
B) Participant observation
C) Laboratory observation
D) Experimental observation

37. Primary focus of descriptive research:

A) Establish causality
B) Explore new phenomena/ new population
C) Describe characteristics of a population/phenomenon
D) Test hypotheses

38. Best test for comparing means of two independent groups:

A) Chi-square
B) t-test
C) ANOVA
D) Correlation

39. Interview with fixed, non-deviating questions:

A) Structured interview
B) Semi-structured interview
C) Unstructured interview
D) Informal interview

40. Key feature of ethnographic research:

A) Short-term data collection
B) Use of experiments
C) In-depth study of cultural groups
D) Statistical analysis

41. Which of the following is an example of a secondary data source?

A) Survey responses
B) Interview transcripts
C) Government reports
D) Experimental results

42. What does 'random assignment' mean in experimental research?

A) Randomly selecting participants from the population
B) Randomly assigning participants to different groups
C) Randomly choosing variables to study
D) Randomly collecting data

43. Which type of validity concerns generalization to real-world settings?

A) Internal validity
B) Construct validity
C) Ecological validity
D) Criterion validity

44. What is a meta-analysis?

A) Analyzing a single study in depth
B) Combining results from multiple studies to draw a general conclusion
C) Reviewing literature on a topic
D) Conducting original research

45. Which is a primary data source?

A) Census data
B) Literature reviews
C) Direct observations
D) Historical records

46. Which type of research aims to generate new theories?

A) Applied research
B) Basic research
C) Experimental research
D) Descriptive research

47. What is the role of an Institutional Review Board (IRB)?

A) To fund research projects
B) To review and approve research involving human subjects
C) To conduct the research
D) To publish findings

48. A common purpose of a focus group is to:

A) Conduct a large-scale survey
B) Gather in-depth qualitative data
C) Perform statistical analysis
D) Manipulate variables in an experiment

49. 'Informed consent' means:

A) Participants' confidentiality keeping procedures and rules
B) Participants agree to take part after being fully informed
C) Researchers' commitment to ethical guidelines
D) Review by an ethics committee

50. The purpose of control variables is to:

A) Test the hypothesis
B) Ensure reliability
C) Eliminate influence of extraneous factors
D) Measure the dependent variable

51. Which method studies the same subjects over a long period?

A) Cross-sectional research
B) Longitudinal research
C) Experimental research
D) Case study research

52. A confounding variable is:

A) Kept constant both independent and dependent variables
B) Influences both independent and dependent variables, leading to spurious effects
C) Manipulated
D) Measured as the outcome

53. What is the primary file format Stata uses for datasets?

A) .csv
B) .xlsx
C) .dta
D) .sav

54. Which R function reads a CSV file?

A) read_csv
B) read.csv
C) csv.read
D) readfile.csv

55. In Stata, which command creates a new variable?

A) gen
B) newvar
C) create
D) addvar

56. In RStudio, which package is commonly used for data manipulation?

A) ggplot2
B) dplyr
C) caret
D) shiny

57. How do you install a package in R?

A) package.install("name")
B) install.packages("name")
C) library("name")
D) install("name")

58. Which Stata command lists all variables in the dataset?

A) variables
B) list
C) describe
D) vars

59. What symbol usually denotes the null hypothesis?

A) H1
B) H2
C) H0
D) Hnull

60. Which R function performs a t-test?

A) t.test()
B) test.t()
C) ttest()
D) test.t

61. In Stata, which command performs a paired t-test?

A) ttest var1, by(var2)
B) ttest var1==var2
C) ttestpaired var1 var2
D) pairtest var1 var2

62. The significance level is typically denoted as:

A) alpha
B) beta
C) p
D) mu

63. If p-value < alpha, you should:

A) Accept the null hypothesis
B) Reject the null hypothesis
C) Accept the alternative hypothesis
D) Fail to reject the null hypothesis

64. Which R function performs a chi-squared test?

A) chi.test()
B) chisq.test()
C) chisquare()
D) chi_square()

65. Which R package is commonly used to create dynamic reports?

A) shiny
B) rmarkdown
C) tidyverse
D) knitr

66. In Stata, which command can export tables to LaTeX?

A) texsave
B) estout
C) outreg2
D) esttab

67. Which function creates a PDF report from an R Markdown file?

A) rmarkdown::render()
B) pdf::render()
C) markdown::pdf()
D) render::rmarkdown()

68. Which format is NOT supported as an R Markdown output document?

A) HTML
B) Word
C) PDF
D) PNG

69. What is the primary purpose of the knitr package?

A) Creating visualizations
B) Data manipulation
C) Dynamic report generation
D) Statistical modeling

70. In Stata, which command creates a table of summary statistics?

A) summarize
B) tabstat
C) descriptive
D) sumtable

71. Which R package is most commonly used for graphics?

A) ggplot2
B) dplyr
C) data.table
D) caret

72. Which Stata command generates a scatter plot?

A) plot
B) scatter
C) graph scatter
D) plot scatter

73. Which R function creates a bar plot?

A) barplot()
B) plot.bar()
C) bar.plot()
D) plotbar()

74. In ggplot2, aes() specifies:

A) Aesthetic mappings
B) Data filters
C) Color schemes
D) Graph titles

75. How do you create a histogram in Stata?

A) histogram varname
B) graph hist varname
C) plot histogram varname
D) hist varname

76. Which R function creates a boxplot?

A) boxplot()
B) plot.box()
C) box.plot()
D) plotbox()

77. What is a Type I error?

A) Rejecting a true null hypothesis
B) Accepting a false null hypothesis
C) Accepting a true null hypothesis
D) Rejecting a false null hypothesis

78. What is a Type II error?

A) Rejecting a true null hypothesis
B) Accepting a false null hypothesis
C) Accepting a true null hypothesis
D) Rejecting a false null hypothesis

79. Having too small a sample size primarily leads to:

A) Increased Type I error rate
B) Decreased power
C) Increased effect size
D) Reduced measurement error

80. Multicollinearity occurs when:

A) Independent variables are too highly correlated
B) The dependent variable is not normal
C) Residuals are heteroscedastic
D) The model is overfitted

81. In Stata, how can you test for heteroscedasticity?

A) hettest
B) test hetero
C) homoscedasticity
D) residualtest

82. Overfitting in regression means the model:

A) Is too complex and fits noise
B) Is too simple
C) Is incorrectly specified
D) Has high variance and low bias

83. In R, which function best checks multicollinearity?

A) cor()
B) vif()
C) multi()
D) collinear()

84. Which Stata command identifies influential data points?

A) influence
B) outlier
C) dfbeta
D) leverage

85. Which R package provides tools for advanced regression modeling?

A) lmtools
B) caret
C) regression
D) advancedreg

86. Which Stata command runs logistic regression?

A) logit
B) reglog
C) logreg
D) logisticreg

87. Which R function fits a linear model?

A) linear.model()
B) lm()
C) fit.model()
D) linmod()

88. Which Stata command merges datasets by keys?

A) append

B) combine

C) merge

D) join

89. In R, how do you merge two data frames by a common column?

A) merge(df1, df2)

B) merge(df1, df2, by = "column")

C) join(df1, df2)

D) merge.dataframes(df1, df2)

90. Which Stata command checks for missing values?

A) misschk

B) misstable

C) checkmiss

D) missing

91. In R, which function provides a basic summary that can reveal NAs?

A) summary.missing()

B) na.summary()

C) summary()

D) missing()

92. Which Stata command generates descriptive statistics?

A) summarize

B) describe

C) stats

D) summary

93. What is the null hypothesis (H0) in hypothesis testing?

A) The hypothesis the researcher is trying to prove

B) There is no effect or no difference

C) The alternative hypothesis

D) A hypothesis that cannot be tested

94. What is the alternative hypothesis (H1)?

A) The hypothesis assumed true initially

B) There is no effect or difference

C) There is an effect or difference

D) Always results in a Type I error

95. A Type I error occurs when:

A) Accepting H0 when false

B) Rejecting H0 when true

C) Accepting H1 when false

D) Rejecting H1 when true

96. The power of a test is the probability of:

A) Rejecting H0 when it is true

B) Rejecting H0 when it is false

C) Accepting H0 when it is false

D) Making a Type I error

97. Which is a parametric test?

A) Chi-square test

B) Mann-Whitney U test

C) t-test

D) Kruskal-Wallis test

98. A z-test is typically used when:

A) Sample sizes are small

B) The population standard deviation is known

C) Data are ordinal

D) Comparing more than two groups

99. What is a critical value?

A) The test statistic value separating rejection from non-rejection regions

B) Max allowable Type II error

C) P(data | H0 true)

D) The mean under H0

100. A two-tailed test evaluates whether the true parameter is:

A) Either larger or smaller than the hypothesized value

B) Only smaller

C) Only larger

D) Equal

101. The significance level (alpha) is the probability of:

A) Making a Type I error

B) Making a Type II error

C) The confidence level

D) The effect size

102. Which test is used for a single proportion (large sample)?

A) Chi-square test

B) ANOVA

C) z-test

D) F-test

103. Comparing means from two related samples requires:

A) Independent samples t-test

B) Paired samples t-test

C) Chi-square test

D) ANOVA

104. An assumption of the classic independent-samples t-test is:

A) Equal variances
B) Unequal variances
C) Zero variances
D) Known variances

105. A Type II error means:

A) Accepting H0 when it is false
B) Rejecting H0 when it is true
C) Accepting H1 when it is true
D) Rejecting H1 when it is true

106. The F-test is used to:

A) Compare two independent means
B) Compare two variances
C) Test independence
D) Compare observed vs expected frequencies

107. Which test compares the variances of two normal populations?

A) Chi-square test
B) t-test
C) F-test
D) ANOVA

108. When testing a correlation coefficient, the usual H0 is that:

A) ρ is significantly positive
B) ρ is significantly negative
C) ρ = 0
D) ρ > 0

109. The purpose of a flow diagram in hypothesis testing is to:

A) Visually represent the steps in the process
B) Calculate the test statistic
C) Determine sample size
D) Illustrate sample distribution

110. A limitation of hypothesis tests is that they:

A) Provide effect size
B) Apply only to large samples
C) Do not measure practical significance
D) Prove hypotheses definitively

111. Which test is typically used to compare more than two means?

A) t-test
B) ANOVA
C) Chi-square test
D) z-test

112. The chi-square test for variance is used to:

A) Determine the population mean

B) Compare sample means

C) Compare a sample variance with a hypothesized population variance

D) Test correlation

113. The chi-square test is considered non-parametric because it:

A) Relies on population parameters

B) Does not require distributional assumptions about the population

C) Compares sample means

D) Uses a normal distribution

114. Applying the chi-square test requires that the data be:

A) Normally distributed

B) Categorical

C) Paired

D) $n < 30$

115. For a chi-square goodness-of-fit test, H0 states that:

A) Observed and expected frequencies differ significantly

B) Observed frequencies follow a specified distribution

C) Variances are equal

D) Sample means are different

116. Yates' continuity correction is used to:

A) Increase the chi-square value

B) Adjust the chi-square value in 2x2 tables with small expected counts

C) Test for normality

D) Adjust for multiple comparisons

117. The chi-square test for independence evaluates whether:

A) Variances are equal

B) Means are different

C) Two categorical variables are related

D) Two samples come from the same population

118. A primary reason to use sampling instead of a census is that sampling:

A) Increases cost

B) Reduces accuracy

C) Is faster and less expensive

D) Eliminates statistical analysis

119. Sampling is NOT appropriate when:

A) The population is too large to study entirely

B) A quick decision is required

C) Every unit must be included

D) Sampling costs less than a census

120. The term for the distribution of a sample statistic over all possible samples is:

A) Sampling distribution
B) Population distribution
C) Sample space
D) Distribution function

121. What is the main purpose of inferential statistics?

A) To describe data
B) To summarize data
C) To infer population characteristics from a sample
D) To collect data

122. Which of the following is true about the t-distribution?

A) Used when sample size is large (n > 30)
B) Used when population sigma is known
C) Has heavier tails than the normal distribution
D) Converges to a binomial distribution

123. The sampling distribution for proportions in large samples is:

A) Normal distribution
B) Chi-square distribution
C) Poisson distribution
D) Exponential distribution

124. According to the Central Limit Theorem, the sampling distribution of the sample mean:

A) Approaches a uniform distribution
B) Approaches a binomial distribution
C) Approaches a normal distribution as n increases
D) Becomes identical to the population distribution

125. A key implication of the Central Limit Theorem is that:

A) The mean of sample means exceeds the population mean
B) The variance of sample means exceeds the population variance
C) The distribution of sample means approaches normality even if the population is not normal
D) The sample mean always equals the population mean

126. Simple random sampling means:

A) Every population member has an equal chance of selection
B) Samples are chosen by specific characteristics
C) Samples are taken at fixed intervals
D) Samples are pre-grouped then selected

127. Stratified sampling means:

A) Population divided into clusters and one cluster sampled
B) Every member has an equal chance of selection
C) Population divided into strata and random samples taken from each

D) Samples taken at regular intervals

128. Sandler's A-test assesses:

A) Variance within one sample
B) Difference between two variances
C) Agreement between two means
D) Magnitude of measurement errors in repeated measures

129. An assumption of Sandler's A-test is that:

A) Data are normally distributed
B) Sample sizes are equal
C) Samples are dependent
D) Population variance is known

130. The standard error of the mean measures:

A) Dispersion of data around the sample mean
B) Variability of the sample mean relative to the population mean
C) Variability of individual observations
D) Central tendency of the population

131. Which factor affects the standard error of the mean?

A) Population mean
B) Sample size
C) Population median
D) Sample median

132. A point estimate is:

A) An interval estimate providing a range
B) A single-value estimate of a population parameter
C) Always equal to the population parameter
D) A biased estimate

133. A property of a good estimator is:

A) Bias
B) Efficiency
C) Variability
D) Complexity

134. When estimating a population mean with small n and unknown sigma, use:

A) Normal distribution
B) t-distribution
C) Binomial distribution
D) Poisson distribution

135. In the confidence-interval formula xbar +/- t * s/sqrt(n), the t represents:

A) Critical value from the normal distribution
B) Critical value from the t-distribution
C) Sample mean
D) Standard deviation

136. The sample proportion phat is:

A) Always equal to the population proportion
B) A point estimate of the population proportion
C) Always biased
D) Constant across samples

137. In the margin-of-error formula +/- z*sqrt(p(1-p)/n), the z represents:

A) Sample proportion
B) Standard deviation
C) Critical value from the normal distribution
D) Mean of sample proportions

138. Which factor does not directly affect required sample size?

A) Desired margin of error
B) Confidence level
C) Population size
D) Population variance

139. When computing sample size using precision rate and confidence level, which is not required?

A) Desired confidence level
B) Population size
C) Precision rate (margin of error)
D) Population variance

140. To obtain a narrower confidence interval at the same confidence level, sample size should be:

A) Increased
B) Decreased
C) Kept constant
D) Doubled

141. Bayesian sample-size determination incorporates:

A) Prior and posterior distributions
B) Confidence intervals only
C) Sample mean alone
D) Sample proportion alone

142. A characteristic of Bayesian sample-size determination is:

A) It ignores prior information
B) Uses maximum-likelihood estimates only
C) Incorporates prior beliefs and evidence
D) Independent of the prior distribution

143. For a skewed population, the CLT generally holds if n >=:

A) 10
B) 20

C) 30
D) 50

144. The main advantage of stratified sampling over simple random sampling is:

A) Easier to administer
B) Reduces sampling error by representing all subgroups
C) Requires fewer resources
D) Simpler analysis

145. The first step in the research process is:

A) Reviewing literature
B) Defining the research problem
C) Collecting data
D) Formulating a hypothesis

146. A research problem should be:

A) Complex and unstructured
B) Clear and precise
C) Vague and general
D) Theoretical only

147. In medical research, informed consent is essential because it:

A) Ensures researcher control
B) Protects legal interests
C) Respects participants' autonomy and rights
D) Simplifies procedures

148. Which ethical principle means "do no harm"?

A) Beneficence
B) Non-maleficence
C) Justice
D) Autonomy

149. Which research design establishes cause-and-effect relationships?

A) Descriptive
B) Experimental
C) Correlational
D) Exploratory

150. A case study is an example of:

A) Quantitative research
B) Qualitative research
C) Experimental research
D) Survey research

151. The primary goal of sampling is:

A) To reduce cost

B) To survey all population members

C) To generalize findings from a sample to a population

D) To simplify data entry

152. Which sampling method ensures equal selection probability?

A) Stratified sampling

B) Cluster sampling

C) Systematic sampling

D) Simple random sampling

153. Which measurement scale classifies data into distinct categories with no order?

A) Ordinal

B) Nominal

C) Interval

D) Ratio

154. An example of an interval scale is:

A) Temperature in degrees C

B) Height in cm

C) Number of children

D) Gender

155. Which data-collection method is typical for qualitative research?

A) Surveys

B) Interviews

C) Questionnaires

D) Checklists

156. Secondary data are:

A) Collected firsthand by the researcher

B) Obtained from existing sources

C) Collected through experiments

D) Gathered via surveys

157. Transforming raw data into meaningful information is called:

A) Data cleaning

B) Data processing

C) Data collection

D) Data storage

158. A common data-processing step is:

A) Data collection

B) Data entry

C) Data analysis

D) Dissemination

159. A sampling frame is:

A) A storage device for samples
B) A list of all items in the population
C) A selection method
D) A non-random technique

160. Dividing the population into subgroups and sampling each defines:

A) Simple random sampling
B) Cluster sampling
C) Stratified sampling
D) Systematic sampling

161. Which test compares means of two independent groups?

A) Chi-square test
B) t-test
C) ANOVA
D) Mann-Whitney U test

162. A p-value < 0.05 indicates:

A) Strong evidence for H0
B) Strong evidence against H0
C) No evidence against H0
D) Data error

163. The chi-square test is used to:

A) Compare means
B) Test independence between categorical variables
C) Compare variances
D) Analyze variance

164. The chi-square test applies when data are:

A) Ordinal
B) Nominal
C) Interval
D) Ratio

165. ANOVA is used to:

A) Compare two means
B) Compare three or more means
C) Compare two variances
D) Compare medians

166. Which technique adjusts means for covariates?

A) ANOVA
B) ANCOVA
C) MANOVA
D) MANCOVA

167. Which non-parametric test compares two related samples?

A) Wilcoxon signed-rank test
B) Mann-Whitney U test
C) Kruskal-Wallis test
D) Friedman test

168. The Mann-Whitney U test compares:

A) Means of two independent samples
B) Medians of two independent samples
C) Means of two related samples
D) Medians of two related samples

169. Which technique identifies underlying factors among correlated variables?

A) Factor analysis
B) Cluster analysis
C) Discriminant analysis
D) Regression analysis

170. Principal Component Analysis (PCA) is primarily used to:

A) Classify data
B) Reduce data dimensionality
C) Predict outcomes
D) Test hypotheses

171. Which section of a research report presents main findings?

A) Introduction
B) Literature review
C) Results
D) Discussion

172. A well-written research report should be:

A) Lengthy and complex
B) Concise and clear
C) Filled with jargon
D) Only positive

173. The primary role of computers in research is:

A) To replace researchers
B) To assist in data collection and analysis
C) To complicate research
D) To eliminate all errors

174. Which software is widely used for statistical analysis?

A) Microsoft Word
B) Adobe Photoshop
C) SPSS
D) AutoCAD

175. Which Stata command summarizes data?

A) summarize
B) tabulate
C) regress
D) correlate

176. In Excel, which function calculates the mean of a range?

A) MEDIAN
B) MODE
C) AVERAGE
D) SUM

177. The main objective of medical research is:

A) To confirm personal opinions
B) To generate new knowledge and improve health outcomes
C) To repeat old experiments
D) To promote treatments

178. Research design is best defined as:

A) Random data collection
B) A detailed plan for conducting the study
C) A hypothesis statement
D) Literature review

179. In research, a variable is:

A) A constant value
B) A characteristic that can vary
C) An unmeasurable factor
D) A controlled setting

180. Which is NOT a step of the scientific method?

A) Defining the problem
B) Data collection
C) Guessing without analysis
D) Drawing conclusions

181. The purpose of a conceptual framework is:

A) To visualize relationships between variables
B) To replace statistics
C) To summarize results
D) To prepare ethics approval

182. Which type of research applies results directly in practice?

A) Basic research
B) Applied research
C) Theoretical research
D) Exploratory research

183. Which of the following is a quantitative variable?

A) Blood-pressure value
B) Blood-group type
C) Marital status
D) Country of origin

184. Which measurement scale has a true zero point?

A) Nominal
B) Ordinal
C) Interval
D) Ratio

185. The primary purpose of pilot studies is:

A) To finalize conclusions
B) To test tools and feasibility before the main study
C) To increase sample size
D) To replace data collection

186. Which sampling technique uses a fixed interval between elements?

A) Simple random sampling
B) Systematic sampling
C) Cluster sampling
D) Snowball sampling

187. An extraneous variable is:

A) Controlled by the researcher
B) Influences the dependent variable but is not of interest
C) The main study variable
D) A constant factor

188. Data coding means:

A) Confusing results
B) Converting raw data into numerical/symbolic form
C) Hiding errors
D) Summarizing theories

189. Data accuracy during entry is ensured by:

A) Cross-checking and validation
B) Ignoring missing values
C) Random typing
D) Deleting incomplete surveys

190. A study that follows participants forward in time is:

A) Retrospective study
B) Cross-sectional study
C) Cohort study
D) Case-control study

191. In a case-control study, participants are selected by:

A) Exposure status
B) Outcome or disease status
C) Random assignment
D) Age category

192. Which test compares frequencies between two categorical variables?

A) t-test
B) ANOVA
C) Chi-square test
D) Correlation analysis

193. The p-value represents:

A) Sample mean
B) Probability of obtaining results by chance
C) Number of variables
D) Level of reliability

194. A confidence interval indicates:

A) Range where the population parameter likely lies
B) Fixed standard deviation
C) Variable's significance
D) Title relevance

195. Sampling bias is reduced by:

A) Convenience sampling
B) Random sampling
C) Quota sampling
D) Purposive sampling

196. A research proposal is:

A) Final paper
B) Preliminary plan outlining purpose, methods, and expected outcomes
C) Survey result
D) Statistical test

197. Reliability of a questionnaire can be measured using:

A) Cronbach's alpha
B) Regression coefficient
C) Mean-square error
D) Skewness value

198. The ethical principle of beneficence means:

A) To gain financial benefit
B) To maximize benefits and minimize harm
C) To share data publicly
D) To speed research

199. Which document defines ethical standards for human research?

A) Nuremberg Code
B) Civil Code
C) Criminal Code
D) Education Law

200. The purpose of randomization in experiments is:

A) Assign by preference
B) Ensure comparable groups and reduce bias
C) Shorten the study
D) Increase sample size

201. Which type of error occurs when a false null hypothesis is accepted?

A) Type I error
B) Type II error
C) Sampling error
D) Measurement error

202. What is the role of peer reviewers in publication?

A) To write articles for authors
B) To evaluate the quality and validity of manuscripts
C) To promote journals
D) To manage finances

203. The abbreviation SPSS stands for:

A) Statistical Package for the Social Sciences
B) Systematic Program for Statistical Studies
C) Simple Processing Software System
D) Sample Population Statistical Sheet

204. In Stata, which command is used to import Excel data?

A) read.csv
B) import excel
C) loadfile
D) use excel

205. Which graphical tool shows the distribution of one quantitative variable?

A) Pie chart
B) Histogram
C) Scatter plot
D) Bar chart

206. What is the main function of a research abstract?

A) To present detailed data
B) To summarize the entire study briefly
C) To list references
D) To describe only methodology

207. Which referencing style is used most often in biomedical journals?

A) APA
B) Vancouver
C) Chicago
D) MLA

208. What is the main difference between qualitative and quantitative research?

A) Qualitative focuses on numbers; quantitative focuses on meanings
B) Qualitative focuses on meanings; quantitative focuses on numbers
C) Both focus on statistical inference
D) There is no difference

209. Which research method emphasizes understanding participants' experiences?

A) Experimental study
B) Qualitative interview
C) Cross-sectional survey
D) Regression analysis

210. In research, what does validity mean?

A) Consistency of measurement
B) Accuracy of measurement
C) Randomness of measurement
D) Complexity of instrument

211. What is a dependent variable?

A) The variable that is manipulated
B) The variable that is affected by changes in another variable
C) A constant factor
D) A random element

212. What is the purpose of inclusion criteria in research?

A) To exclude participants
B) To define which participants qualify to enter the study
C) To randomize data
D) To collect secondary data

213. The process of selecting participants who meet certain characteristics is called:

A) Randomization
B) Sampling
C) Recruitment
D) Stratification

214. Which sampling technique selects groups rather than individuals?

A) Simple random sampling
B) Cluster sampling
C) Stratified sampling
D) Systematic sampling

215. What is a research hypothesis?

A) A proven fact

B) A tentative explanation or prediction that can be tested

C) A random assumption

D) A statistical method

216. What is the main purpose of data analysis?

A) To collect data

B) To organize, interpret, and draw conclusions from data

C) To delete outliers

D) To visualize results only

217. Which type of study examines data from different groups at one point in time?

A) Cohort study

B) Cross-sectional study

C) Experimental study

D) Longitudinal study

218. What is a research gap?

A) An unanswered or underexplored question in existing studies

B) A statistical error

C) A missing data point

D) A mistake in sampling

219. The process of checking the accuracy and consistency of collected data is called:

A) Validation

B) Classification

C) Tabulation

D) Quantification

220. In research, triangulation is used to:

A) Test normality of data

B) Use multiple sources or methods to confirm findings

C) Measure reliability

D) Simplify analysis

221. What is the main purpose of literature citation?

A) To show reading habits

B) To give credit to original authors and avoid plagiarism

C) To fill space

D) To support personal opinions

222. What is bias in research?

A) Random variation in data

B) Systematic error that affects validity

C) Measurement error due to instruments

D) Missing data in the dataset

223. What does placebo mean in a clinical trial?

A) An inactive substance used to compare with a treatment
B) A strong experimental drug
C) A harmful chemical
D) An independent variable

224. In medical ethics, autonomy refers to:

A) The researcher's independence
B) The participant's right to make informed decisions
C) The doctor's authority
D) Institutional control

225. What is the purpose of the Declaration of Helsinki?

A) To define national research funding
B) To regulate ethical principles for medical research involving humans
C) To measure validity
D) To approve budgets

226. What does confidentiality mean in research ethics?

A) Sharing data with all colleagues
B) Protecting participants' personal information
C) Publishing all names
D) Allowing sponsors to see raw data

227. Which document is typically submitted to obtain ethical approval?

A) Research proposal
B) Informed consent form
C) Ethics application with full protocol
D) Literature review

228. Difference between primary and secondary data:

A) Primary is collected directly; secondary from existing sources
B) Primary is old; secondary is new
C) Primary is from books; secondary from surveys
D) Primary is theoretical; secondary practical

229. Which statistical test analyzes the relationship between two continuous variables?

A) Chi-square test
B) Correlation test
C) ANOVA
D) Mann-Whitney U test

230. Standard deviation is:

A) The square root of variance
B) The average of all data
C) The total number of cases
D) Difference between means

231. A research instrument is reliable if it:

A) Produces consistent results under the same conditions
B) Always gives higher values
C) Measures multiple variables simultaneously
D) Involves random errors

232. Data visualization is used to:

A) Improve text structure
B) Present data clearly via graphs and charts
C) Delete missing values
D) Perform correlation

233. Which graph is best for showing proportions?

A) Bar chart
B) Pie chart
C) Histogram
D) Line chart

234. The role of sample-size calculation is:

A) To guarantee adequate statistical power for detecting meaningful effects
B) To minimize the time required to perform analytical procedures in research
C) To intentionally introduce bias during participant or data selection processes
D) To make scientific manuscript writing simpler without methodological rigor

235. The function of the SPSS Output Viewer is:

A) To generate complete narrative reports without human interpretation
B) To display outcomes of statistical analyses for review and interpretation
C) To digitally collect and store completed survey questionnaires from respondents
D) To perform structured coding of raw dataset values prior to input processing

236. In Stata, the command summarize varname provides:

A) A visual graph illustrating distribution patterns across variables
B) Descriptive statistical indicators related to the selected variable
C) A generated output containing multi-level regression model results
D) A comprehensive matrix examining correlation among multiple variables

237. An advantage of using Stata over Excel in data analysis is:

A) The ability to format tables and visuals more easily for presentation purposes
B) The availability of advanced statistical tools with high analysis reproducibility
C) Integration of automated literature review from scientific databases
D) Inclusion of a built-in module for composing detailed research manuscripts

238. What is a manuscript cover letter?

A) A standardized template used exclusively for manual data entry into software
B) A formal letter submitted to the editor to introduce and justify the article
C) An evaluative feedback document provided by a peer reviewer during assessment
D) An informed consent form signed by participants taking part in the investigation

239. RCT stands for:

A) Random Case Trial
B) Randomized Controlled Trial
C) Retrospective Clinical Test
D) Repeated Control Testing

240. Main feature of a double-blind study:

A) Both researcher and participant know treatment assignment
B) Only participants know
C) Neither researcher nor participant knows
D) Both groups use placebos

241. Which of the following is an example of descriptive statistics?

A) Mean, median, mode
B) Regression analysis
C) ANOVA
D) Chi-square test

242. The interquartile range (IQR) measures:

A) The spread of the middle 50% of data
B) The mean difference
C) The total data range
D) The mode of a dataset

243. Main goal of data cleaning is:

A) To remove errors and inconsistencies from data
B) To increase dataset size
C) To generate hypotheses
D) To simplify graphs

244. The null hypothesis (H0) states that:

A) There is no significant relationship or difference
B) There is a strong relationship
C) The results are biased
D) The experiment is invalid

245. The alternative hypothesis (H1) suggests that:

A) There is no effect
B) There is an effect or relationship
C) Variables are identical
D) Data are invalid

246. A research population is:

A) The entire group relevant to the study
B) A tested subset
C) A list of data points
D) A statistical summary

247. Random sampling is used to:

A) Minimize selection bias
B) Increase the number of variables
C) Test only volunteers
D) Simplify analysis

248. A cross-over design is:

A) A study where each participant receives several interventions sequentially
B) A research method based solely on observational data collection techniques
C) An experimental approach applying only a double-blind placebo methodology
D) A statistical process used to combine results from multiple published studies

249. Data tabulation means:

A) The systematic process of organizing information into structured tables
B) The removal of all missing values from the original research database
C) Application of statistical hypothesis testing to confirm assumed results
D) Classification of randomly observed variables with minimal organization

250. Informed consent includes:

A) Providing research details, risks, benefits, and participant legal rights
B) Sharing financial compensation and payment information with respondents
C) Presenting formal academic qualifications of the principal investigator
D) Stating deadlines and official time limits related to study implementation

251. A main advantage of standardized questionnaires is:

A) Ensuring comparability of collected results across different research studies
B) Increasing the likelihood of excluding ethical oversight requirements
C) Eliminating the necessity of obtaining participant informed consent
D) Enhancing the amount of unintended bias introduced during data collection

252. In ANOVA, the F-statistic is the ratio of:

A) Two unrelated sample variances that are computed independently
B) Variance between groups divided by variance within groups
C) Arithmetic mean values calculated from two distinct populations
D) A covariance measurement illustrating variable interdependence

253. Data normalization is:

A) Scaling values to a common range without affecting meaningful differences
B) Eliminating every detected outlier from the dataset without further review
C) Rounding each data point to a reduced number of decimal places
D) Transforming quantitative data values into categorical (qualitative) form

254. The median is:

A) The observation that occurs most frequently within a given dataset
B) The central value when all data are arranged in ascending or descending order
C) The smallest numerical value identified in the collection of observations
D) The total sum of values divided by the count of data points collected

255. Standard error (SE) indicates:

A) The variability of the sample mean compared to the true population mean
B) The numerical difference between calculated median and mode values
C) The level of error attributed to imperfections in the measurement tool
D) The directional bias present during selection of research sample participants

256. Which test is non-parametric?

A) ANOVA
B) Mann-Whitney U test
C) t-test
D) z-test

257. Multicollinearity in regression means:

A) Independent variables are highly correlated
B) Dependent variable is missing
C) Residuals are random
D) Outliers are removed

258. Heteroscedasticity is:

A) Unequal variance of residuals
B) Equal variance across groups
C) Perfect correlation
D) Missing-variable bias

259. To assess the strength of a linear relationship, use:

A) Mean
B) Correlation coefficient (r)
C) Standard deviation
D) Median

260. $p < 0.05$ usually indicates:

A) Results due to chance
B) Statistically significant results
C) Invalid results
D) Sample too small

261. The best chart for showing time trends is a:

A) Pie chart
B) Bar chart
C) Line chart
D) Histogram

262. Best research method for rare diseases:

A) Cross-sectional study
B) Case-control study
C) Cohort study
D) Experimental design

263. External validity means:

A) Generalizability to other populations or settings
B) Accuracy of measurement tools
C) Random error control
D) Internal consistency

264. A data-coding sheet contains:

A) Variables, labels, and assigned numerical codes
B) Names of researchers
C) Research timeline
D) Interview transcripts

265. A scatter plot is used for:

A) Displaying the relationship between two quantitative variables
B) Comparing the frequencies of multiple study groups
C) Showing proportional distributions of categorical data
D) Demonstrating the ranked order of observed outcomes

266. A control variable is:

A) A variable intentionally kept constant to avoid influencing the results
B) A variable that is actively manipulated by the researcher during the study
C) A variable that changes randomly without systematic control
D) A variable that depends entirely on other measured variables

267. An example of secondary data is:

A) Data collected from medical records
B) Data collected by a survey
C) Lab test results
D) Patient observations

268. The purpose of research ethics committees is:

A) To approve financial resources for research projects
B) To verify that studies comply with ethical and safety regulations
C) To conduct statistical examination of collected study results
D) To carry out routine administrative data processing activities

269. In epidemiology, incidence refers to:

A) The complete number of already existing disease cases in a population
B) The count of newly occurring disease cases within a defined time
C) The total number of individuals who are considered at health risk
D) The measurement of symptom frequency across different subjects

270. Prevalence means:

A) The total number of new disease cases occurring during a study period
B) The complete number of existing cases at a specific time point
C) The total mortality rate observed in the studied population
D) The total number of clinically recovered patients reported

271. Meta-analysis is:

A) A structured review statistically combining results from several studies
B) A research project based on a single observational investigation
C) A qualitative data collection technique using guided interviewing
D) A descriptive summary report without statistical interpretation

272. Publication bias is:

A) The tendency to publish mainly studies with statistically significant outcomes
B) A random computational mistake occurring during data analysis procedures
C) The use of incorrect citation practices in preparing the manuscript
D) A time delay caused by reviewers during the evaluation process

273. Gray literature refers to:

A) Non-commercial or unpublished research such as theses or institutional reports
B) Peer-reviewed scientific articles featured in highly ranked journals
C) Promotional marketing material intended for public advertisement
D) Research posters presented solely at scientific conference events

274. A prospective study:

A) Monitors selected individuals moving forward in time to observe outcomes
B) Investigates previously recorded information from hospital documentation
C) Reviews and assesses already published scientific literature
D) Utilizes group discussion sessions to collect qualitative perspectives

275. A retrospective study:

A) Collects data going forward in time
B) Analyzes existing or past data
C) Conducts lab experiments
D) Predicts future results

276. The main goal of a systematic review is:

A) To summarize and synthesize results from multiple studies objectively
B) To conduct interviews
C) To create new experiments
D) To publish personal opinions

277. An operational definition is:

A) A precise and measurable description of how a variable will be assessed
B) A general written summary of literature sourced from scientific databases
C) A broad theoretical framework explaining abstract study concepts
D) A simple graphical diagram representing study components visually

278. Pilot testing of instruments ensures:

A) Practical feasibility, measurement reliability, and clarity of data tools
B) Automatic participant allocation through a randomization procedure
C) Immediate ethical approval from relevant institutional review authority
D) Achievement of statistical significance for all measured study outcomes

279. Data triangulation is:

A) The use of several independent data sources to confirm research findings
B) Construction of exactly three analytical variables within the data
C) Repeated utilization of a single questionnaire tool without modifications
D) Manual application of random allocation techniques during experimentation

280. Construct validity is:

A) The extent to which an instrument accurately measures the intended concept
B) The practical capability to reproduce identical research results repeatedly
C) The degree of variability observed across multiple study outcome values
D) The accuracy level of input during manual data entry processes

281. Which statistical test examines relationships between categorical variables?

A) Pearson correlation
B) Chi-square test
C) t-test
D) Regression analysis

282. Which test compares three or more means?

A) ANOVA
B) t-test
C) Chi-square
D) Correlation

283. Regression analysis aims to determine:

A) The relationship between dependent and independent variables
B) The average of data points
C) The validity of questionnaires
D) The proportion of missing data

284. Graphically, the normal distribution appears as a:

A) Bell-shaped curve
B) Skewed curve
C) Flat line
D) Zigzag pattern

285. Skewness indicates:

A) The asymmetry of a data distribution
B) The variability of variance
C) The number of observations
D) The mean value

286. Flattening or peaking of a distribution refers to:

A) Kurtosis
B) Variance
C) Dispersion
D) Normality

287. Confidence level means:

A) The probability that the interval contains the true population parameter
B) The total number of samples included in the current investigation
C) The correlation coefficient calculated from the study variables
D) The numeric value of the statistical test statistic used in analysis

288. A common confidence level in research is:

A) Fifty percent of the measurement accuracy reported in analysis
B) Ninety percent of the commonly accepted statistical certainty level
C) Ninety-five percent reliability used in most scientific publications
D) Seventy-five percent reliability applied for preliminary assessments

289. Effect size is:

A) The magnitude of difference or association observed between variables
B) The arithmetic mean derived from the collected sample data
C) The calculated probability of committing statistical error in testing
D) The measured variance computed for the available research sample

290. A data entry error is:

A) Incorrect recording of data values during the input phase
B) Absence of selected respondents from the participation process
C) Partial execution of required literature review activities
D) Violation of ethical standards during research implementation

291. In clinical trials, an adverse event refers to:

A) Any undesired medical occurrence arising in a study participant
B) A beneficial and expected outcome following an intervention
C) Random computational mistake within statistical analysis software
D) Official rejection of the trial during the ethical review procedure

292. The function of a Data Monitoring Committee (DMC) is:

A) To supervise safety and scientific validity throughout clinical trials
B) To manage participant recruitment and enrollment processes
C) To prepare scientific manuscripts intended for journal submission
D) To collect and label biological samples for laboratory evaluation

293. Dropout rate means:

A) Percentage of participants who discontinue involvement before study end
B) Total number of individuals initially enrolled in the investigation
C) Statistical deviation introduced during the process of sample selection
D) Quantity of research variables excluded from the analytical procedure

294. Attrition bias is:

A) Systematic distortion caused by unequal participant loss between groups
B) Error resulting from faulty random allocation of study participants
C) Bias produced due to incorrect or inconsistent measurement tools
D) Selective publication of studies showing statistically significant results

295. A data confidentiality breach is:

A) Unauthorized disclosure or access to participants' private information
B) Failure to obtain required data from certain study respondents
C) Incorrect application of coding principles during data processing
D) External contamination of biological samples used in analysis

296. Repeating a study to confirm results is called:

A) Verification
B) Replication
C) Validation
D) Randomization

297. The introduction of a research paper should:

A) Present study background, problem statement, and research objectives
B) Provide references with links retrieved directly from Scopus database records
C) Display graphical illustrations without additional explanatory text
D) Offer a brief overview of descriptive statistical outcomes only

298. The methods section of a manuscript should:

A) Explain in detail how the research was structured and executed
B) Present analytical outcomes obtained from data processing
C) Provide a narrative summary of general study interpretations and results
D) Include a complete glossary of abbreviations used throughout the manuscript

299. The results section should primarily contain:

A) Objective factual findings presented clearly without author interpretation
B) Explanatory evaluation of study outcomes compared with earlier research work
C) Citations and references supporting the theoretical background discussion
D) Expressive personal viewpoints of the researcher on study performance

300. The discussion section should:

A) Interpret results, compare with previous studies, and highlight implications
B) Present only unformatted raw numerical data collected during the study
C) Repeat detailed methodology described earlier in the manuscript sections
D) Contain solely tables without explanatory narrative or contextual relevance